



ROBOTS MEET ARTS

EDUCATIONAL ROBOTICS AND CODING MEET
ARTS AND HUMANITIES IN INCLUSIVE LEARNING
ENVIRONMENTS

MODULE 2

TEACHING WITH EDUCATIONAL ROBOTICS AND
CODING

COORDINATOR & PARTNERS



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Theory and key concepts

Introduction

In today's rapidly evolving educational landscape, Educational Robotics (ER) and coding are emerging as transformative tools for fostering 21st-century skills such as critical thinking, creativity, collaboration, and computational thinking. These tools go beyond technical skill development – they create dynamic, interactive, and learner-centered environments that empower students to become active creators rather than passive consumers of technology.

This module explores how educators can effectively integrate ER and coding into their teaching practices to enhance student engagement, motivation, and learning outcomes across subjects. Whether used in STEM disciplines or creatively linked to the arts and humanities, ER and coding provide powerful entry points into problem-solving, storytelling, and real-world application.

Section 1 begins with practical pedagogical strategies that support inclusive and hands-on learning. Later, Section 2 examines how various established teaching methodologies – from project-based and inquiry-based learning to differentiated instruction and learning boxes – can be adapted to suit the unique affordances of ER and coding activities. Throughout, the emphasis is on student agency, collaborative learning, and reflective practice, ensuring that teaching with robotics and coding is both accessible and inspiring for all learners.

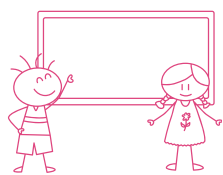
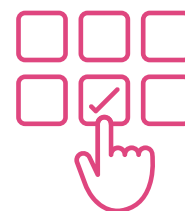


PART 1 -General Tips for Teaching Robotics and Coding

Educational robotics and coding offer much more than technical skills — they open doors to creative expression, collaborative exploration, and real-world problem-solving. To make the most of these opportunities, it is crucial to create a supportive and student-centered learning environment. The following strategies are designed to enhance student learning, motivation, and engagement, while also nurturing key transversal competencies such as resilience, curiosity, and communication.

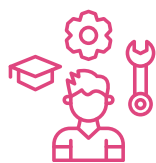
Tips for Teaching Robotics and Coding

Embrace student choice: Allow students to select which practice levels to complete based on their interests and comfort levels. This approach boosts engagement and gives students a sense of ownership over their learning process, making it more meaningful and enjoyable.



Encourage student explanations: Encourage students to present and explain their solutions to the class. This not only reinforces their understanding and problem-solving skills but also fosters leadership and confidence as they articulate their thought processes and methodologies.

Embrace mistakes: Normalize and celebrate mistakes by tracking instances where students debug errors successfully. This cultivates a positive learning environment that values resilience and continuous improvement, helping students to see mistakes as valuable learning opportunities.



Draw real-world connections: Connect coding and robotics tasks to real-world scenarios to enhance relevance and interest. Discuss how robots are used in various industries or how coding powers everyday technology, thereby showing students the practical implications of their learning.

Collaborative projects: Promote teamwork by assigning group projects that require students to collaborate, design, and program together. This collaborative effort enhances problem-solving skills, fosters peer learning, and builds a sense of community within the classroom.





Reflect and iterate: After completing tasks or projects, encourage students to reflect on what worked, what didn't, and how they can improve. Promote an iterative mindset where students continuously refine their solutions, fostering a culture of ongoing learning and adaptation.



Scaffolded learning: Start with simple tasks and gradually increase complexity as students build confidence and skills. Provide scaffolding through guided practice and gradually release responsibility, allowing students to develop their independence and expertise incrementally.

Example in Practice: A Primary Classroom with LEGO SPIKE Kits

In a lively primary school classroom, students are introduced to a hands-on robotics challenge using LEGO SPIKE Essential kits. The teacher presents **two creative options**: students can either build a **delivery robot** that simulates transporting small packages **or a dancing robot** that performs choreographed movements to music. This element of **student choice** boosts excitement and personal investment, as learners form teams and select the project that resonates most with their interests.



Over the next hour, the classroom buzzes with activity. One group designing a delivery robot sketches a route for their robot to follow using tape on the floor, thinking critically about how to program the motors for turns and speed control. Another team working on a dancing robot experiments with block-based coding to synchronize movements with rhythmic beats. **As they test their robots, mistakes naturally arise** – a delivery robot spins in circles instead of moving straight. Rather than feeling discouraged, the students burst into laughter, diagnose the issue, and correct a motor direction command, celebrating their successful debugging. **These moments are encouraged by the teacher, who consistently reminds students that mistakes are part of the learning journey.**



To deepen learning, the teacher initiates a guided **group discussion**, prompting students to think about **real-world applications**: “Where do we see **delivery robots in real life?**” Students mention automated carts in hospitals, robots delivering food in hotels, and even warehouse drones. This real-world connection helps students see the broader purpose behind their coding efforts and understand how technology serves people in diverse contexts. As part of the extension, one group enhances their robot with sound effects and colored sensors. They program the robot to change LED lights depending on delivery status (e.g., red when loaded, green when delivered) and play cheerful sounds upon completing tasks. The teacher provides **scaffolded support**, showing them how to use conditional logic blocks to activate these features.

At the end of the session, students engage in a brief **reflection circle**. They discuss what went well, such as learning to reverse a motor or synchronize dance moves with music, and what they’d improve next time — like stabilizing the robot’s frame or making the route shorter. Some express interest in using sensors for obstacle avoidance in future designs.

In short, this project integrates the tips presented above:

- Student choice drove intrinsic motivation.
- Peer explanations during presentations solidified understanding.
- Mistake-embracing cultivated a safe learning environment
- Real-world relevance sparked curiosity and deeper thinking.
- Collaboration fostered teamwork and shared learning.
- Scaffolding supported progressive mastery.
- Reflection and iteration nurtured metacognition and growth mindset.



PART 2 - Integrating established teaching methodologies in robotics and coding

Educational robotics and coding are rapidly becoming integral parts of modern education, fostering essential skills such as critical thinking, problem-solving, and creativity. To effectively teach these subjects, educators can leverage a variety of well-known teaching methodologies. Each of these methodologies offers unique advantages and can be adapted to enhance the learning experience in robotics and coding. This section explores several prominent teaching methodologies and how they can be applied to educational robotics and coding to create an engaging and effective learning environment.

Project-based learning (PBL)

It is a student-centered methodology that involves students working on a project over an extended period. In educational robotics, PBL can involve designing, building, and programming robots to solve real-world problems. This approach encourages critical thinking, creativity, and practical application of theoretical knowledge.

You can learn more about PBL by watching this [video](#) “Project Based Learning: Explained.” by PBL Works that explains in clear language the essential elements of Project Based Learning (PBL). Also, this [video](#) “How to Design Project-Based Learning Activities” by European Schoolnet Academy provides you with a preliminary checklist and 7 basic steps for designing a Project Based Learning activity to develop your students’ competences.

Example 1 - Project-Based Activity for Primary School



Smart Home Automation with Robots

Students are working on a smart home model that is equipped with simple sensors and smart devices. Students are programming and testing the robot to control the home automation system as lights and sounds, etc. (picture generated with DALL-E)

Example 2 - Project-Based Activity for Primary School



A day at the farm

Students are working on a digital story on Scratch. They are programming the characters of the farm to speak, move, and interact with each other.

(picture generated with DALL-E)

Inquiry-Based Learning (IBL)

Inquiry-Based Learning is a student-centered approach where learning is driven by curiosity, questioning, and active investigation. In the context of educational robotics and coding, it transforms the classroom into a space of discovery, where students become investigators, designers, and problem-solvers, rather than passive recipients of information.

At the heart of IBL is the questioning process. Instead of starting with instructions, students are presented with a problem or phenomenon – such as, “How can we program a robot to help sort recyclable materials?” or “What sensors can help a robot detect obstacles?” From there, students generate hypotheses, test their ideas through building and coding, and refine their understanding based on the results. Teachers serve as facilitators, guiding the process with open-ended questions, providing scaffolding when needed, and encouraging reflection.

IBL encourages a hands-on, minds-on approach. Students might:

- Observe how a robot behaves under certain conditions.
- Ask why the robot isn't completing a task correctly.
- Research how similar problems are solved in real-world contexts.
- Test different coding sequences or mechanical configurations.
- Document findings and propose improvements.

For **example**, in a unit on environmental sustainability, students may be challenged to design a robotic prototype that can help clean a simulated waterway. They must ask what kinds of sensors or parts are needed, test their robot's ability to detect and pick up debris, and iterate based on trial outcomes. This type of learning builds critical thinking, perseverance, and a scientific mindset, making concepts more meaningful and transferable.

Example of an Inquiry-Based Learning Activity: Designing a Waterway-Cleaning Robot

In this inquiry-driven robotics activity, primary students are engaged as young researchers and engineers, challenged to design a robotic prototype that helps clean polluted waterways. The project integrates robotics, coding, and environmental education, but it is the students' questions and curiosity that shape the learning journey.

The process begins with an **open-ended exploration of the problem**: “How can robots help solve real-world environmental issues like water pollution?” Students conduct **research** on existing robotic solutions, observe how different robots function, and **pose their own questions about design, materials, and programming needs**.

Guided by these questions, they **investigate possible solutions, formulate hypotheses, and collaboratively design prototypes** using tools like LEGO SPIKE kits. They make critical decisions about which sensors, attachments, or coding blocks to use based on their identified problem and desired functionality.

As students test their robots in simulated environments, they continuously analyze outcomes, troubleshoot issues like sensor misreadings or mechanical faults, and refine their designs through iteration. Their problem-solving is rooted in inquiry: Why did the robot fail to collect small debris? What code can improve accuracy? How do sensor adjustments affect performance?

Throughout, the teacher supports as a facilitator of inquiry, prompting students to reflect, rethink, and build on each learning experience. In the final phase, students present their solutions, explain their design rationale, and connect their learning to broader environmental concerns — reinforcing the idea that inquiry is not only about finding answers, but about developing a mindset of exploration, critical thinking, and continuous improvement.



Collaborative Learning

Collaborative Learning is an educational approach that emphasizes group work, shared goals, and collective responsibility for learning. In robotics and coding, it is particularly effective because these fields naturally involve problem-solving in teams, dividing tasks, and integrating different perspectives. In a collaborative learning environment, students are placed in small, diverse groups where they co-construct knowledge and negotiate solutions. A typical robotics project may assign roles such as engineer, coder, project manager, and documentarian, ensuring that each student contributes in a meaningful way while also learning from others. Key elements of effective collaborative learning in robotics include:

- **Shared responsibility:** Each student has a role but the success of the project depends on team cooperation.
- **Peer teaching:** Students explain concepts and strategies to one another, reinforcing their own understanding.
- **Dialogue and decision-making:** Students learn to communicate clearly, resolve conflicts, and justify their choices.
- **Collective reflection:** Teams evaluate their process and outcomes together.

Example of Collaborative Learning with Tale-Bot and Animal Storytelling



In this collaborative learning activity, young students (ages 6–8) work in teams to create an interactive story using Tale-Bot, a programmable robot designed for early coding education. The theme, “A Day in the Forest,” invites children to design settings, create animal characters, and program Tale-Bot’s movements and voice lines as it embarks on a journey to deliver a message with the help of forest animals.

Each student is responsible for a different part of the story, promoting teamwork, shared decision-making, and creativity. Students collaborate to develop the plot, assign roles, build scenery, and debug Tale-Bot’s behavior when things go wrong. They perform their story segments sequentially, forming a complete classroom narrative that celebrates collective effort. This activity enhances communication skills, creativity, basic programming, and empathy, while making learning joyful, social, and meaningful through hands-on robotics and storytelling.

Learning boxes

Learning boxes are inspired by challenge-based education and promote hands-on, project-oriented learning. Each learning box is centered around a specific challenge, integrating multiple subjects to provide a holistic educational experience. Tasks are scaffolded to increase in complexity, encouraging skill development and confidence.

Key features include collaborative learning, where students work in groups to enhance teamwork and communication skills, and reflective practice, allowing students to consider their learning process and apply new knowledge to different contexts. The implementation involves introducing the challenge, exploration and research, practical application, iteration and improvement, and presentation and reflection. The learning boxes can connect robotics and coding to real-world challenges and boost student motivation and interest. For example, a learning box on climate change could include a "Rescue Robot Adventure" task where students design, build, and program a simple robot to navigate a pretend disaster area, locate objects, and deliver essential supplies.

Here you can find the [presentation of the Learning boxes methodology](#) and [testimonials](#) from teachers and students.

Example of a Learning box activity - Rescue Robot Adventure



A group of students is engaged in a learning box activity. The students are actively opening and examining the contents, which include instruction sheets, materials, tools, robot components and task cards. The box is climate change and includes several relevant tasks. One of the key tasks in the learning box is the Rescue Robot Adventure.

The Rescue Robot Adventure involves designing, building, and programming a robot to rescue a character from a natural disaster, simulating a real-world rescue mission. The activity begins with an introduction to the challenge, through a scenario brief and discussion among students. Students then brainstorm ideas, sketch their robot designs, and select appropriate materials and tools from the learning box. Then, students assemble their robot, test their functionality, and make necessary adjustments. They write and debug the code to control the robot, ensuring it can detect obstacles and navigate the terrain effectively. The task includes trial runs to fine-tune the robot's performance. After the mission, students reflect on the challenges faced and the lessons learned.



Differentiated Learning Methodology

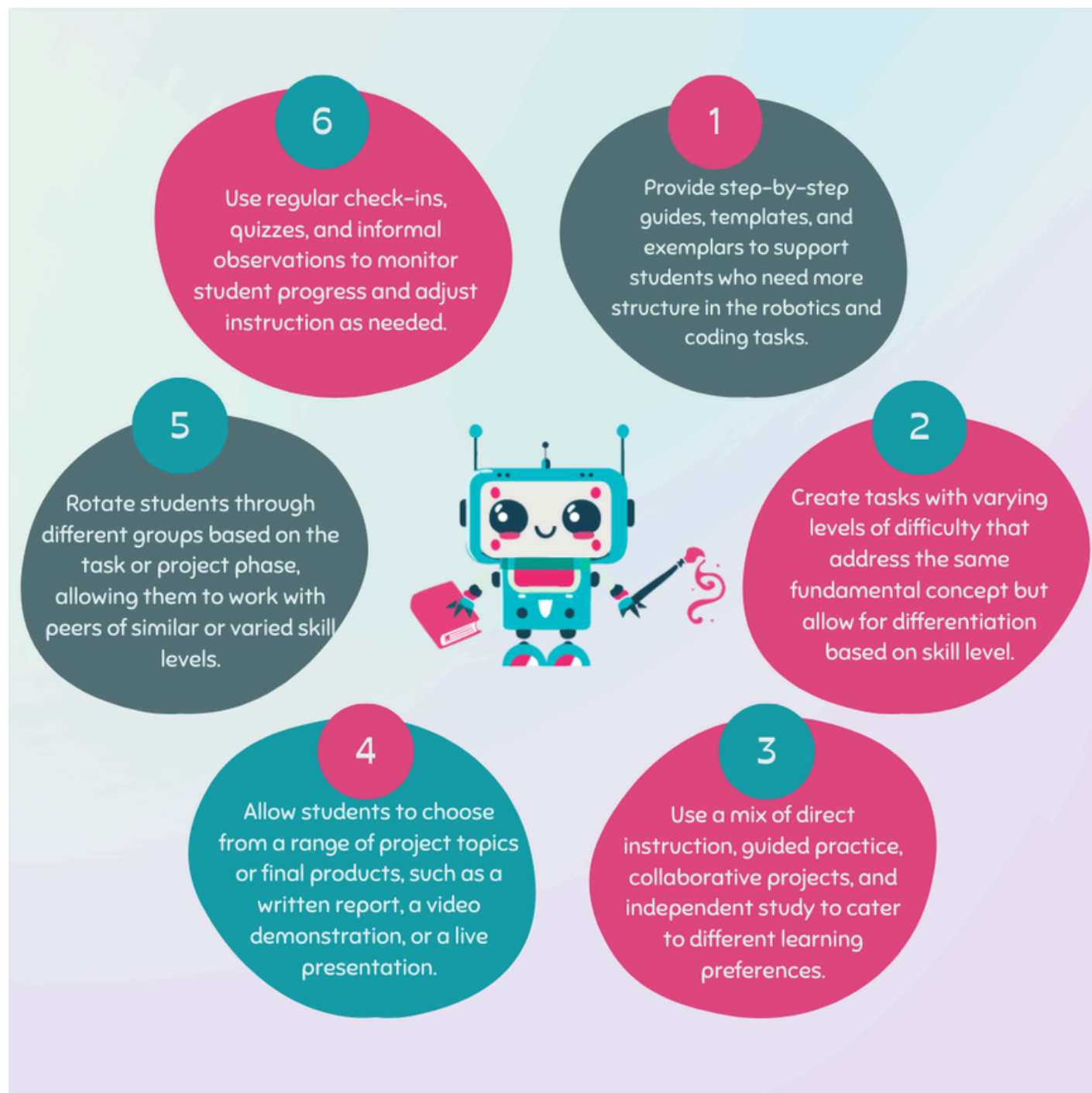
This methodology aims to address the diverse learning needs, preferences, and abilities of students within a classroom. This methodology involves tailoring teaching strategies, content, resources, and assessments to ensure that all students can achieve their learning objectives.

In the context of educational robotics and coding, differentiated learning can play a crucial role in engaging students and enhancing their understanding of complex concepts. First of all, the content of the classes can be adapted based on students' readiness, interests, and learning profiles and the teacher can provide different materials, multimedia resources, or hands-on activities tailored to varying skill levels. In addition, the teaching process can be differentiated by incorporating various strategies. For example, some students might benefit from direct instruction on how to write code, while others might prefer to learn through hands-on self-paced experimentation with robotics kits. Group projects can be designed to encourage collaboration among students with different skill levels, allowing them to learn from each other. Additionally, using technology tools such as coding simulators or online tutorials can cater to individual learning paces.

In a differentiated robotics and coding class, students might start with a common project brief, such as designing a robot to complete a specific task. Based on their skill levels, different groups might work on varying aspects of the project. Beginners could focus on assembling the robot and learning basic coding commands, while advanced students might tackle more complex programming challenges and optimize the robot's performance. Throughout the project, the teacher provides individualized support and resources tailored to each group's needs. By employing differentiated learning in educational robotics and coding, teachers can ensure that all students are challenged appropriately and supported in their learning journey. This approach not only caters to diverse learning needs but also fosters a more inclusive and engaging educational environment.



Tips for applying Differentiated Learning Methodology when using Educational robotics and coding





Examples of differentiated robotics and coding activities

Below are concrete, short examples of differentiated robotics and coding activities designed to promote equity, diversity, and inclusion, especially for engaging underrepresented groups (e.g. girls in STEM, students with disabilities, migrant or multilingual learners, and students with varying socioeconomic backgrounds).

1. Culturally Relevant Robot Stories

Target: Multilingual learners, migrant students

Activity: Students program a robot (e.g., Bee-Bot or Tale-Bot) to follow a path that tells a story from their home culture or country. They use voice recording features to narrate parts of the story in their native language and English.

Differentiation: Students choose the complexity of the route and language used; visuals support those with limited language skills.

2. Girls in Tech – Role Model Challenge

Target: Girls and students underrepresented in STEM

Activity: Students research women innovators in technology and robotics, then build a robot to represent a challenge that figure solved (e.g., a robot astronaut, rescue robot, etc.).

Differentiation: Students choose whether to focus on storytelling, design, or programming. Those with strong visual or verbal skills can lead presentation; coders can focus on the logic.

3. Accessibility Design Hack

Target: Students with and without disabilities

Activity: Students design a robot that helps people with a disability (e.g., a robotic arm that delivers items, or a color-detecting robot for someone with color blindness).

Differentiation: Tasks vary by ability: some build and wire the robot, others write simple code or prepare an explanatory video using assistive tech.

Inclusive Outcome: Encourages empathy and universal design thinking.

4. Multi-Level Maze Challenge

Target: Mixed-ability groups

Activity: All students receive the same robot maze challenge, but tasks vary by difficulty:

- Beginners: Move the robot forward, backward, and turn.
- Intermediate: Add loops or conditions.
- Advanced: Use variables and sensors.
- Differentiation: Students self-select challenge levels or are gently guided; peer coaching is encouraged.



5. Low-Cost Robotics Using Recycled Materials

Target: Schools with limited resources or low-SES communities

Activity: Students build robot prototypes using recycled household materials (e.g., cardboard, bottle caps), combined with basic electronics or microcontrollers like Micro:bit or Makey Makey.

Differentiation: Focus can be on design, storytelling, or coding depending on student comfort and access.



PART 3 - Integrating Robotics in Arts and Humanities: Extended Guidance for Teachers

Many teachers feel uncertain about how to use educational robotics beyond STEM disciplines. Yet, with creative planning and pedagogical flexibility, robotics can powerfully enhance teaching in arts, literature, languages, history, civic education, and cultural studies. This section provides practical guidance for integrating robots into arts and humanities lessons, along with realistic classroom examples that emphasize inquiry, creativity, and cross-disciplinary learning.

Storytelling and Literature through Robotics

Storytelling offers one of the most natural entry points for using robots in humanities. In a language or literature class, students can write a short story, folk tale, or myth and then bring it to life using programmable robots such as Tale-Bot, Bee-Bot, or LEGO SPIKE. For example, they can design a story map where the robot becomes a moving character. With simple directional coding and voice recordings, the robot navigates through different plot points while narrating the story aloud. This approach deepens students' comprehension and promotes both oral and written expression, especially for multilingual learners.

Exploring History with Robotic Timelines

Robotics can also support historical learning by helping students reconstruct key events and simulate the journeys of historical figures. Teachers might ask students to create a floor-based timeline where robots move along a path representing important events, such as the civil rights movement or ancient trade routes. As the robot arrives at each milestone, students can trigger recorded narrations or act out mini-presentations. This method makes abstract historical narratives concrete and spatial, improving recall and engagement.



Drama, Dialogue, and Robotic Theatre

In performing arts or language classes, students can use robots as animated puppets or theatrical characters. After writing short dialogues or scripts, students program robots to act out scenes with movement and voice. They may add costumes, backgrounds, or props to enrich the performance. For example, in a modern fairy tale project, each robot represents a character who interacts with others according to a programmed sequence of speech and motion. This format supports oral language skills, expressive movement, and team-based creativity.

Cultural and Geographical Discovery

Robots can act as “travelers” or “guides” in geography and cultural studies. Teachers can facilitate a unit where students research different countries or regions and program a robot to visit key locations on a world map. At each destination, students share facts about language, traditions, or festivals. In one variation, students add audio recordings in multiple languages, turning the robot into a multilingual cultural ambassador. This activity supports both intercultural awareness and global citizenship education.

Ethics and Moral Exploration

Robots can also be used in social studies, ethics, or philosophy classes to simulate moral dilemmas and encourage discussion. For example, students can design a scenario where a robot must choose between helping a friend or reaching a goal, with consequences triggered by its programmed path. After watching the outcome, students engage in guided reflection about values, choices, and social responsibility. This kind of experiential learning brings ethical thinking into action, fostering both empathy and critical reasoning.

Pedagogical Tips for Implementation

To ensure success, begin with small, manageable projects that align with existing curriculum goals. Co-designing activities with students helps build ownership and creativity. Teachers should feel free to adapt the complexity of coding and design tasks to match the age group and confidence level of learners. Integrating storytelling, visual design, sound, and movement allows students with different strengths and learning styles to contribute meaningfully.

Robotics in the arts and humanities doesn’t require technical expertise—it requires a willingness to explore, collaborate, and experiment. With clear guidance, teachers can unlock robotics’ full potential to enrich subject-specific learning in inclusive and imaginative ways.